PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 10: Binomial GLMs, Binomial GLMMs, Quasi-binomial and Beta-binomial Models

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Motivating example: bumblebee infection

Define aggregated responses

$$S_i = \sum_{j=1}^{n_i} Y_{ij}$$

- i indexes colonies
- j indexes bees within colony i
- Y_{ij} is a binary indicator (1 = infected, 0 = not infected)
- n_i is the number of observed bees in colony i at the end of experiment

1. Binomial generalized linear model (GLM)

$$S_i \sim \text{Binomial}(n_i, p_i),$$
$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

- S_i: number of infected bees in colony i
- n_i: total observed bees in that colony
- p_i: probability of infection for a bee in colony i
- Binomial(n_i, p_i): binomial distribution with n_i trials and success probability p_i.
- g: a link function
- $\boldsymbol{x}_i$: the vector of covariates (including the intercept and various indicators).
- $\boldsymbol{\beta}$: the vector of fixed effects.

Accordingly,

$$p_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta}), \qquad E(S_i) = n_i p_i, \qquad \text{Var}(S_i) = n_i p_i (1 - p_i).$$

With the binomial specification, individual bee outcomes are independent conditional p_i. This can be implausible for bees sharing a colony. An aggregated binomial likelihood does not automatically account for extra variation due to within-colony dependence.

Available link functions for binomial GLMs

The link function g connecting the probability p_i to the linear predictor $\boldsymbol{x}_i^\top \boldsymbol{\beta}$. Changing the link changes how covariates relate to p_i, but does not by itself account for overdispersion or within-colony dependence. Logistic regression is the binomial GLM with the logit link; other links are possible.

Link	Mathematical form $g(p)$	R specification
Logit (default)	$\log\{p/(1-p)\}$	`binomial(link = "logit")`
Probit	$\Phi^{-1}(p)$	`binomial(link = "probit")`
Log	$\log(p)$	`binomial(link = "log")`
Complementary log-log	$\log\{-\log(1-p)\}$	`binomial(link = "cloglog")`
Cauchit	$\tan\{\pi(p-1/2)\}$	`binomial(link = "cauchit")`

2. Binomial generalized linear mixed-effects model (GLMM)

A GLMM adds random effects to GLM to allow dependence across colonies:

$$S_i \mid \boldsymbol{\gamma}_i \sim \text{Binomial}(n_i, p_i),$$
$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \boldsymbol{z}_i^\top \boldsymbol{\gamma}_i, \qquad \boldsymbol{\gamma}_i \overset{\text{iid}}{\sim} \text{MVN}(0, \Sigma_\gamma).$$

- $\boldsymbol{z}_i$: the vector of covariates corresponding to random effects.
- $\boldsymbol{\gamma}$: the vector of random effects.

3. Quasi-binomial model

A quasi-binomial model adjusting variance without specifying a new model for p_i. Specifically, the quasi-binomial GLM is

$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}, \qquad E(S_i) = n_i p_i, \qquad \text{Var}(S_i) = \phi n_i p_i (1-p_i);$$

The quasi-binomial GLMM is

$$g(p_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \boldsymbol{z}_i^\top \boldsymbol{\gamma}_i, \qquad E(S_i \mid \boldsymbol{\gamma}_i) = n_i p_i, \qquad \text{Var}(S_i \mid \boldsymbol{\gamma}_i) = \phi n_i p_i (1-p_i), \qquad \boldsymbol{\gamma}_i \overset{\text{iid}}{\sim} \text{MVN}(0, \Sigma_\gamma).$$

- ϕ is an estimated dispersion factor. For ϕ greater than 1, quasi-binomial standard errors are typically larger than ordinary binomial standard errors.
- The quasi-binomial GLM may be fitted via `R` function `glm` with `family = quasibinomial(link = "logit")`.
- The quasi-binomial GLMM may be fitted via `R` function `MASS::glmmPQL` with `family = quasibinomial(link = "logit")`.

4. Beta-binomial model

$$P_i \sim \text{Beta}\left(\mu_i \frac{1-\rho}{\rho}, (1-\mu_i)\frac{1-\rho}{\rho}\right),$$
$$S_i \mid P_i \sim \text{Binomial}(n_i, P_i),$$
$$\text{logit}(\mu_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

Or, equivalently,

$$S_i \sim \text{BetaBinomial}(n_i, \mu_i, \rho),$$
$$\text{logit}(\mu_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}.$$

Accordingly,

$$E(S_i) = n_i \mu_i,$$
$$\text{Var}(S_i) = n_i \mu_i (1-\mu_i)\{1 + (n_i - 1)\rho\}.$$

- P_i: the random infection probability for colony i
- μ_i: mean infection probability for colony i.

- $\rho \in (0, 1)$: the within-colony correlation. Unlike a constant quasi-binomial dispersion, the beta-binomial variance inflation depends on colony size n_i.
- Model fitting may be implemented via `R` function `glmmTMB::glmmTMB` with `family = betabinomial(link = "logit")`.